similar set for his system, which we can label τ, so we don't mix them up with Alice's. Alice may have several σ operators, and likewise Bob may have several τ operators. With this framework in hand, we're ready to explore composite systems in greater depth. Later on, in Lecture 7, we'll explain how to work with tensor product operators in component form—expressed as matrices and column vectors.

By now, there should be no doubt in your mind that quantum physics is different from classical physics, right down to its logical roots. In this lecture and the next one, I am going to hit you even harder with this idea. We are going to discuss an aspect of quantum physics that is so different from classical physics that, as of this writing, it has puzzled—and aggravated—physicists and philosophers for almost 80 years. It drove its discoverer, Einstein, to the conclusion that something very deep is missing from quantum mechanics, and physicists have been arguing about it ever since. As Einstein realized, in accepting quantum mechanics, we are buying into a view of reality that is radically different from the classical view.

6.2 CLASSICAL CORRELATION

Before we get to quantum entanglement, let's spend a few minutes on what we might call classical entanglement. In the following experiment, Alice (A) and Bob (B) will get some help from Charlie (C).

Charlie has two coins in his hands—a penny and a dime. He mixes them up and holds them out, one in each hand, to Alice and Bob, and gives one coin to each of them. No one looks at the coins and no one knows who has which. Then, Alice gets on the shuttle to Alpha Centauri while Bob stays in Palo Alto. Charlie has done his job and doesn't matter anymore (sorry, Charlie).

Before Alice's big trip, Alice and Bob synchronize their clocks—they have done their relativity homework and accounted for time dilation and all that. They agree that Alice will look at her coin just a second or two before Bob looks at his.